

Problem 1a) (15 points)

Given the image region as shown in Figure 1 and $V = \{1\}$, what is the shortest m-path between p (the pixel at the upper-left corner) and q (the pixel at the bottom-right corner)? Give the length of the path if the path exists.

```

1 0 1 1 0
0 1 0 1 0
0 0 1 1 0
1 1 0 1 0
0 1 1 1 1

```

Step 1: Identify pixels in V

p

1	0	1	1	0
0	1	0	1	0
0	0	1	1	0
1	1	0	1	0
0	1	1	1	1

q

All pixels with value 1 are highlighted. These are the only candidates for the m-path since $V = \{1\}$.

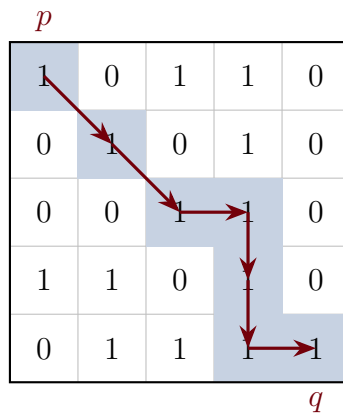
Step 2: Apply m-adjacency rules

Two pixels with values in V are **m-adjacent** if:

- They are 4-adjacent (share an edge), OR
- They are diagonally adjacent AND their shared 4-neighbors contain no pixels with values in V .

The second condition prevents diagonal shortcuts when a 4-connected bridge exists, eliminating path ambiguity.

Step 3: Trace the shortest m-path



Path: $(0, 0) \rightarrow (1, 1) \rightarrow (2, 2) \rightarrow (2, 3) \rightarrow (3, 3) \rightarrow (4, 3) \rightarrow (4, 4)$

Diagonal move justifications:

- $(0, 0) \rightarrow (1, 1)$: Shared 4-neighbors are $(0, 1) = 0$ and $(1, 0) = 0$. Neither is in V , so m-adjacent.
- $(1, 1) \rightarrow (2, 2)$: Shared 4-neighbors are $(1, 2) = 0$ and $(2, 1) = 0$. Neither is in V , so m-adjacent.

Results

The shortest m-path exists and has **length 6**.

Problem 1b) (15 points)

To take a picture for fireworks at night, give at least two operations that should be applied to improve the visualization quality of the image and explain.

Step 1: Understand the Scene Characteristics

Fireworks at night present specific challenges:

- **High dynamic range:** Very bright fireworks against a very dark sky
- **Desired appearance:** Pure black sky with vivid, crisp firework sparks
- **No hidden information:** The dark sky contains nothing meaningful to reveal



Original fireworks image

Step 2: Brightening Operations

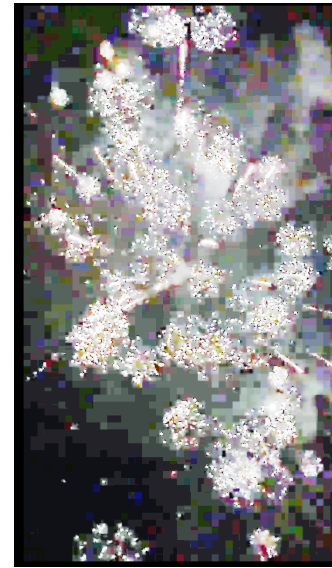
These operations attempt to reveal detail in shadows by lifting dark intensities.



Gamma ($\gamma = 0.5$)



Log Transform

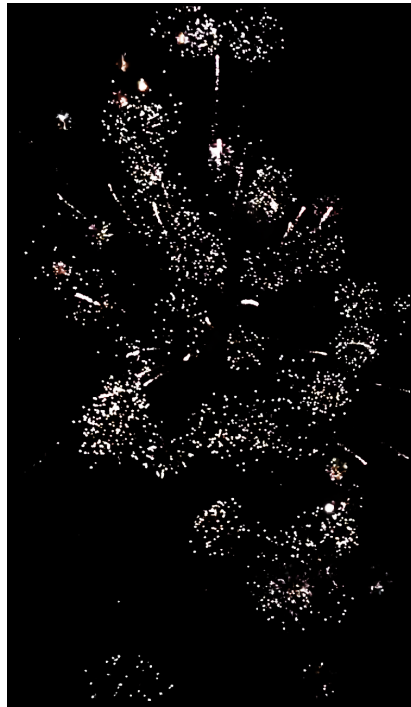


Histogram Equalization

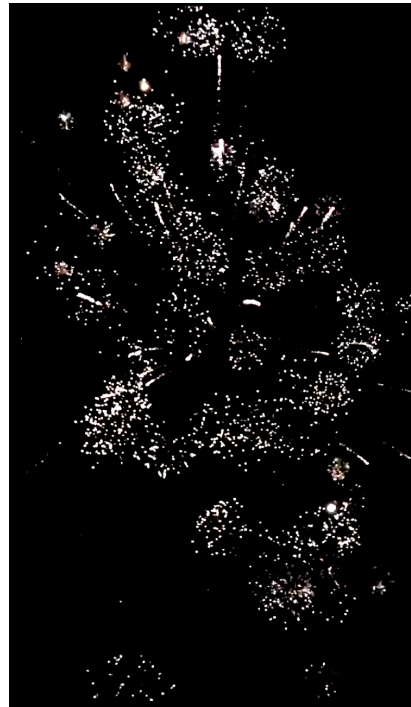
Effect: The black sky becomes gray, revealing noise and sensor artifacts. This is **undesirable** for fireworks because the sky *should* be black—there is no meaningful information to reveal.

Step 3: Contrast Enhancement Operations

These operations push dark values toward black and light values toward white.



S-Curve Contrast



Boosted Contrast

Effect: The sky becomes pure black while the fireworks remain bright. This is **desirable** for aesthetic night photography.

Step 4: Sharpening Operations

These operations enhance edges and fine details.



Unsharp Masking



High-Boost Filtering

Effect: Firework sparks appear crisper with more defined edges.

Step 5: Color Enhancement Operations



Saturation Boost



Warm Temperature



Warm + Color Contrast

Saturation: Uniformly boosts all color intensity—can cause clipping in already-vivid areas.

Warm Temperature: Shifts colors toward orange/yellow by boosting red and reducing blue channels, enhancing the fiery appearance.

Warm + Color Contrast: Applies warmth first, then an S-curve in Lab color space to push warm colors warmer and cool colors cooler, creating rich golden tones.

Step 6: Consider the Optimization Goal

The “correct” operations depend entirely on what you are trying to achieve:

Goal	Recommended Operations	Avoid
Aesthetic (fireworks)	S-curve, sharpening, saturation	Gamma, Hist EQ
Surveillance	Gamma, log transform, Hist EQ	S-curve
Medical imaging	CLAHE, careful contrast	Extreme transforms
Object detection	Thresholding, edge detection	Aesthetic enhancements

The same operation can be *beneficial* or *harmful* depending on the goal. Histogram equalization is excellent for revealing hidden details in surveillance footage, but terrible for fireworks photography where the dark background should remain dark.

Step 7: Consider the Display Medium

The target display device also affects which operations are most beneficial:

Display	Prioritize	Less Important
Mobile phone (small screen)	Color, contrast, brightness	Fine sharpening
Desktop monitor	Sharpness, fine detail	Can be subtle
Print	Resolution, color accuracy	Different gamma curve

On a mobile phone, viewers cannot resolve fine detail, so aggressive sharpening provides little benefit—but punchy colors and contrast help images “pop” on a small screen viewed in variable lighting. On a desktop monitor, fine details are visible, making sharpening more valuable.

Results

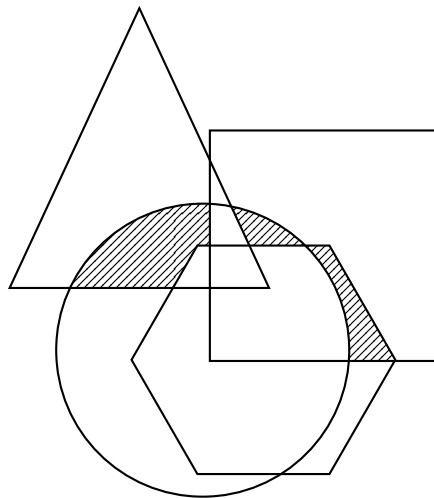
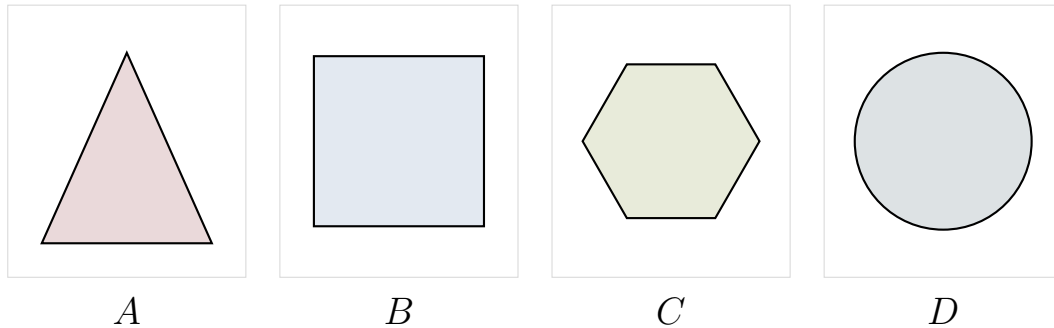
Recommended operations for fireworks photography:

- S-curve contrast enhancement:** Apply a sigmoid transformation $s = \frac{1}{1+e^{-k(r-0.5)}}$ to push dark pixels darker and bright pixels brighter. This ensures a pure black sky while maintaining bright firework sparks.
- Saturation boost** (especially for mobile display): Increase color saturation to make the firework colors more vivid and visually striking.
- Sharpening** (especially for desktop/print): Apply unsharp masking or high-boost filtering to enhance the crispness of firework spark edges.

Operations to avoid: Gamma correction ($\gamma < 1$), log transforms, and histogram equalization—these lift the dark sky and reveal noise, degrading the aesthetic quality.

Problem 2) (20 points)

A, B, C and D are four sets. Give expressions for the set shown shaded in Figure 2 in terms of A, B, C, and D. Note that all of the shaded areas constitute of one set

**Step 1: Identify the Shaded Regions**

Examining the diagram, the shaded area consists of three distinct regions where exactly two shapes overlap, excluding the areas where three or more shapes would overlap:

- The region where *A* (triangle) and *D* (circle) overlap, but not *B* or *C*
- The region where *B* (rectangle) and *D* (circle) overlap, but not *A* or *C*
- The region where *B* (rectangle) and *C* (hexagon) overlap, but not *A* or *D*

Step 2: Write Set Expressions for Each Region

Using set intersection ($\cap$) and complement notation (X^c):

Region 1: Points in both A and D , but not in B or C :

$$A \cap D \cap B^c \cap C^c$$

Region 2: Points in both B and D , but not in A or C :

$$B \cap D \cap A^c \cap C^c$$

Region 3: Points in both B and C , but not in A or D :

$$B \cap C \cap A^c \cap D^c$$

Step 3: Combine Using Union

The complete shaded region is the union of all three regions.

Results

The shaded set can be expressed as:

$$(A \cap D \cap B^c \cap C^c) \cup (B \cap D \cap A^c \cap C^c) \cup (B \cap C \cap A^c \cap D^c)$$

Problem 3) (25 points)

Given a 3x3 spatial filter $\frac{1}{9} \begin{bmatrix} -1 & -1 & -1 \\ -1 & -17 & -1 \\ -1 & -1 & -1 \end{bmatrix}$ answer the following questions:

- a) (10 points) What type of filter is it? When applying this filter on an image, what kind of effect you will expect in the output image? Why? (Hint: is this filter an image smoothing filter, an image sharpening filter or an edge detector?)

- b) (15 points) Perform the image convolution using this filter on a 3x3 image $\begin{bmatrix} 3 & 5 & 12 \\ 7 & 18 & 11 \\ 5 & 9 & 15 \end{bmatrix}$.

Give the CROPPED filtering result for the output image. (Hint: you need to assume an appropriate zero mapping)

Step 1: Analyze the Filter Type

The filter $\frac{1}{9} \begin{bmatrix} -1 & -1 & -1 \\ -1 & -17 & -1 \\ -1 & -1 & -1 \end{bmatrix}$ can be decomposed as:

$$= -\frac{1}{9} \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix} - \frac{16}{9} \begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix} = -(\text{averaging}) - \frac{16}{9}(\text{identity})$$

This is a **sharpening filter** (high-boost filter) with inversion:

- The center weight magnitude (17) is much larger than neighbor weights (1), emphasizing the center pixel and enhancing edges
- All negative coefficients mean the output is inverted (negative image)
- Sum of coefficients $= \frac{1}{9}(-8 - 17) = -\frac{25}{9}$, so constant regions are scaled by $-\frac{25}{9} \approx -2.78$

Step 2: Apply Zero Padding

Pad the image with zeros around the border:

$$\begin{bmatrix} 0 & 0 & 0 & 0 & 0 \\ 0 & 3 & 5 & 12 & 0 \\ 0 & 7 & 18 & 11 & 0 \\ 0 & 5 & 9 & 15 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

Step 3: Compute Convolution — Row 0

For each output pixel, multiply the filter by the corresponding 3×3 neighborhood and sum. The filter coefficients are -1 for all neighbors and -17 for the center, with a normalization factor of $\frac{1}{9}$.

Position (0,0): Neighborhood $\begin{bmatrix} 0 & 0 & 0 \\ 0 & 3 & 5 \\ 0 & 7 & 18 \end{bmatrix}$

$$\frac{1}{9}[(-1)(0 + 0 + 0 + 0 + 5 + 0 + 7 + 18) + (-17)(3)] = \frac{1}{9}[-30 - 51] = \frac{-81}{9} = -9$$

Position (0,1): Neighborhood $\begin{bmatrix} 0 & 0 & 0 \\ 3 & 5 & 12 \\ 7 & 18 & 11 \end{bmatrix}$

$$\frac{1}{9}[(-1)(0 + 0 + 0 + 3 + 12 + 7 + 18 + 11) + (-17)(5)] = \frac{1}{9}[-51 - 85] = \frac{-136}{9} \approx -15.1$$

Position (0,2): Neighborhood $\begin{bmatrix} 0 & 0 & 0 \\ 5 & 12 & 0 \\ 18 & 11 & 0 \end{bmatrix}$

$$\frac{1}{9}[(-1)(0 + 0 + 0 + 5 + 0 + 18 + 11 + 0) + (-17)(12)] = \frac{1}{9}[-34 - 204] = \frac{-238}{9} \approx -26.4$$

Step 4: Compute Convolution — Row 1

Position (1,0): Neighborhood $\begin{bmatrix} 0 & 3 & 5 \\ 0 & 7 & 18 \\ 0 & 5 & 9 \end{bmatrix}$

$$\frac{1}{9}[(-1)(0 + 3 + 5 + 0 + 18 + 0 + 5 + 9) + (-17)(7)] = \frac{1}{9}[-40 - 119] = \frac{-159}{9} \approx -17.7$$

Position (1,1): Neighborhood $\begin{bmatrix} 3 & 5 & 12 \\ 7 & 18 & 11 \\ 5 & 9 & 15 \end{bmatrix}$

$$\frac{1}{9}[(-1)(3 + 5 + 12 + 7 + 11 + 5 + 9 + 15) + (-17)(18)] = \frac{1}{9}[-67 - 306] = \frac{-373}{9} \approx -41.4$$

Position (1,2): Neighborhood $\begin{bmatrix} 5 & 12 & 0 \\ 18 & 11 & 0 \\ 9 & 15 & 0 \end{bmatrix}$

$$\frac{1}{9}[(-1)(5 + 12 + 0 + 18 + 0 + 9 + 15 + 0) + (-17)(11)] = \frac{1}{9}[-59 - 187] = \frac{-246}{9} \approx -27.3$$

Step 5: Compute Convolution — Row 2

Position (2,0): Neighborhood $\begin{bmatrix} 0 & 7 & 18 \\ 0 & 5 & 9 \\ 0 & 0 & 0 \end{bmatrix}$

$$\frac{1}{9}[(-1)(0 + 7 + 18 + 0 + 9 + 0 + 0 + 0) + (-17)(5)] = \frac{1}{9}[-34 - 85] = \frac{-119}{9} \approx -13.2$$

Position (2,1): Neighborhood $\begin{bmatrix} 7 & 18 & 11 \\ 5 & 9 & 15 \\ 0 & 0 & 0 \end{bmatrix}$

$$\frac{1}{9}[(-1)(7 + 18 + 11 + 5 + 15 + 0 + 0 + 0) + (-17)(9)] = \frac{1}{9}[-56 - 153] = \frac{-209}{9} \approx -23.2$$

Position (2,2): Neighborhood $\begin{bmatrix} 18 & 11 & 0 \\ 9 & 15 & 0 \\ 0 & 0 & 0 \end{bmatrix}$

$$\frac{1}{9}[(-1)(18 + 11 + 0 + 9 + 0 + 0 + 0 + 0) + (-17)(15)] = \frac{1}{9}[-38 - 255] = \frac{-293}{9} \approx -32.6$$

Results

Part a): This is a **sharpening filter** (high-boost filter) with inversion. It enhances edges by heavily weighting the center pixel relative to its neighbors, but produces a negative (inverted) output due to all negative coefficients.

Part b): The cropped filtering result with zero padding is:

$$\frac{1}{9} \begin{bmatrix} -81 & -136 & -238 \\ -159 & -373 & -246 \\ -119 & -209 & -293 \end{bmatrix} = \begin{bmatrix} -9 & -15.1 & -26.4 \\ -17.7 & -41.4 & -27.3 \\ -13.2 & -23.2 & -32.6 \end{bmatrix}$$

The negative values confirm the filter produces an inverted output.

Problem 4) (25 points)

An image with intensities in the range $r \in [0, L - 1]$ has the pdf

$$p_r(r) = \begin{cases} \frac{4r}{3(L-1)^2} & 0 \leq r \leq L-1 \\ 0 & \text{otherwise} \end{cases}$$

where $L - 1$ is the maximum intensity value. It is desired to transform the intensity levels of this image so that they will have the specified pdf

$$p_z(z) = \begin{cases} \frac{2z}{(L-1)^2} & 0 \leq z \leq L-1 \\ 0 & \text{otherwise} \end{cases}$$

Assume that the intensity value is continuous and find the transformation function $z = T(r)$ in terms of r and z that will accomplish this.

(Hints: The transformation function of histogram equalization is given as

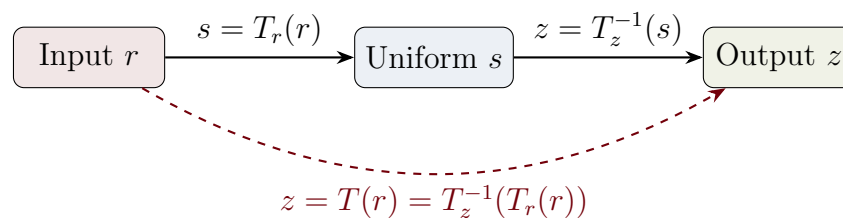
$$s = T(r) = (L-1) \int_0^r p_r(w) dw$$

)

Step 1: Understand the Goal

We want to transform pixel intensities from distribution $p_r(r)$ to distribution $p_z(z)$. The challenge is that there is no direct formula to go from one arbitrary distribution to another.

Key insight: Histogram equalization maps *any* distribution to a uniform distribution. We can use this as a “bridge” between the two distributions.



Since both r and z map to the *same* uniform distribution s , the values of s must be equal. This lets us solve for z in terms of r .

Step 2: Equalize the Input Distribution $p_r(r)$

Apply histogram equalization to the input. The CDF (cumulative distribution function) gives us s :

$$\begin{aligned} s = T_r(r) &= (L-1) \int_0^r p_r(w) dw \\ &= (L-1) \int_0^r \frac{4w}{3(L-1)^2} dw \\ &= (L-1) \cdot \frac{4}{3(L-1)^2} \cdot \frac{w^2}{2} \Big|_0^r \\ &= (L-1) \cdot \frac{4}{3(L-1)^2} \cdot \frac{r^2}{2} = \boxed{\frac{2r^2}{3(L-1)}} \end{aligned}$$

Step 3: Equalize the Desired Distribution $p_z(z)$

Apply the same process to the desired output distribution:

$$\begin{aligned} s = T_z(z) &= (L-1) \int_0^z p_z(w) dw \\ &= (L-1) \int_0^z \frac{2w}{(L-1)^2} dw \\ &= (L-1) \cdot \frac{2}{(L-1)^2} \cdot \frac{w^2}{2} \Big|_0^z \\ &= (L-1) \cdot \frac{2}{(L-1)^2} \cdot \frac{z^2}{2} = \boxed{\frac{z^2}{L-1}} \end{aligned}$$

Step 4: Set Equal and Solve for z

Both expressions equal the same uniform value s . Setting them equal:

$$\underbrace{\frac{2r^2}{3(L-1)}}_{T_r(r)} = \underbrace{\frac{z^2}{L-1}}_{T_z(z)}$$

Multiply both sides by $(L-1)$:

$$\frac{2r^2}{3} = z^2$$

Take the square root (using positive root since intensities are non-negative):

$$z = \sqrt{\frac{2}{3}} r = \frac{\sqrt{6}}{3} r$$

Results

The transformation function is:

$$z = T(r) = \sqrt{\frac{2}{3}} r = \frac{\sqrt{6}}{3} r \approx 0.816 r$$

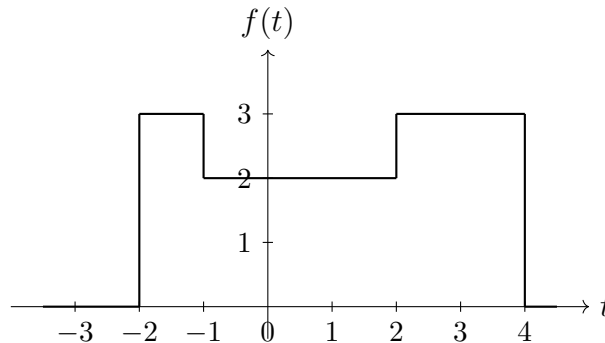
Interpretation: This is a simple linear scaling that compresses intensities by about 18%. Every input intensity r maps to a slightly smaller output intensity $z \approx 0.816r$.

Why does this work? The input PDF $p_r(r) = \frac{4r}{3(L-1)^2}$ rises more steeply than the desired PDF $p_z(z) = \frac{2z}{(L-1)^2}$. To “spread out” the distribution, we compress the intensity values, which reduces the slope of the resulting histogram.

Bonus Question (20 points)

Perform the Fourier transform to a function as shown in Figure 3.

(Hint: the definition of the Fourier transform of a function is $F(u) = \int_{-\infty}^{\infty} f(t)e^{-j2\pi ut} dt$)

**Step 1: Express the Function Piecewise**

From the figure, the function $f(t)$ is:

$$f(t) = \begin{cases} 3 & -2 \leq t < -1 \\ 2 & -1 \leq t < 2 \\ 3 & 2 \leq t < 4 \\ 0 & \text{otherwise} \end{cases}$$

Step 2: Key Formula for Rectangular Pulse

For a constant h over interval $[a, b]$, the Fourier transform is:

$$\int_a^b h e^{-j2\pi ut} dt = \frac{h}{j2\pi u} (e^{-j2\pi ua} - e^{-j2\pi ub})$$

Cheat sheet version: Just plug in height h , left bound a , right bound b . No integration needed.

Step 3: Apply the Formula to Each Segment**Segment 1:** $h = 3, a = -2, b = -1$

$$F_1(u) = \frac{3}{j2\pi u} (e^{-j2\pi u(-2)} - e^{-j2\pi u(-1)}) = \frac{3}{j2\pi u} (e^{j4\pi u} - e^{j2\pi u})$$

Segment 2: $h = 2, a = -1, b = 2$

$$F_2(u) = \frac{2}{j2\pi u} (e^{-j2\pi u(-1)} - e^{-j2\pi u(2)}) = \frac{2}{j2\pi u} (e^{j2\pi u} - e^{-j4\pi u})$$

Segment 3: $h = 3, a = 2, b = 4$

$$F_3(u) = \frac{3}{j2\pi u} (e^{-j2\pi u(2)} - e^{-j2\pi u(4)}) = \frac{3}{j2\pi u} (e^{-j4\pi u} - e^{-j8\pi u})$$

Step 4: Combine and SimplifySum all three contributions and factor out $\frac{1}{j2\pi u}$:

$$\begin{aligned} F(u) &= \frac{1}{j2\pi u} [3e^{j4\pi u} - 3e^{j2\pi u} + 2e^{j2\pi u} - 2e^{-j4\pi u} + 3e^{-j4\pi u} - 3e^{-j8\pi u}] \\ &= \frac{1}{j2\pi u} [3e^{j4\pi u} + (-3 + 2)e^{j2\pi u} + (-2 + 3)e^{-j4\pi u} - 3e^{-j8\pi u}] \\ &= \frac{1}{j2\pi u} [3e^{j4\pi u} - e^{j2\pi u} + e^{-j4\pi u} - 3e^{-j8\pi u}] \end{aligned}$$

Results

The Fourier transform is:

$$F(u) = \frac{1}{j2\pi u} [3e^{j4\pi u} - e^{j2\pi u} + e^{-j4\pi u} - 3e^{-j8\pi u}], \quad u \neq 0$$

For $u = 0$, compute the area directly:

$$F(0) = \int_{-\infty}^{\infty} f(t) dt = 3(1) + 2(3) + 3(2) = 15$$